One Additional Observation, More Conditional Certificates: A Paired Study in a Finite System Class

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Abstract

Does a development-selected query policy certify more future targets than a fixed query at the same environmental observation budget? We compare two deterministic policies on 5,000 pre-specified paired draws from a finite frame of system classes and histories. A certificate requires unanimous target predictions in a nonempty compatible candidate set. The selected policy certifies 372 targets versus 283 for the fixed query: 90 selected-only and one fixed-only outcome. The net difference is 1.78 percentage points, with the pre-specified at-least-95% interval [1.23, 2.29]. All 655 issued certificates match the privileged audit. Equal observation budgets require unequal local computation; 92.56% of selected-policy draws remain uncertified. Coverage of the true system by the candidate class is a premise. This is a conditional acquisition result, not a new active-learning principle, live LLM result, universal guarantee or proof of a gain of at least two percentage points.

1 Question, contract and findings

This study asks whether *acquiring* one further legal observation helps resolve an uncertainty: a future target outcome left ambiguous by the available history. It is a separate experiment with deterministic actors, a covering class of deterministic systems with at most three states and binary inputs and outputs, and two permitted queries. A conditional target certificate is issued only when every candidate consistent with the history and the observed query response predicts the same target. Class coverage is an explicit premise of this certificate. From the eight-input history's cutoff, q_0 applies input 0 and q_1 applies input 1; each returns only that transition's output. The target is the pair of second outputs on input sequences $(0, 0)$ and $(1, 0)$, each starting at the original cutoff. Appendix A defines the state and output conventions.

The fixed policy always chooses the first query. The development-selected policy maximizes the number of nonempty response branches having a single target outcome, breaks ties by minimizing the largest feasible branch's number of distinct targets, and finally chooses the first query. Branch and candidate counts are not calibrated probabilities. Both arms receive exactly one observation on every draw, including initially unambiguous and duplicate records. Separate processes and state copies prevent cross-arm reuse. Neither actor receives the true system's identity, private state or target answer. All predictions were committed before a separate privileged target audit.

The protocol fixed the policy, tie rules, $N = 5{,}000$, confidence procedure and validity gate before the main draw. These are local, hash-bound precommitments, not an independently timestamped public preregistration. The hypothesis class covers at most three states; the sampled worlds instead have exactly three minimal reachable states, after development exclusions. Sampling was uniform with replacement over 8,372 eligible behavior classes crossed with 256 input histories: 2,143,232 class–history pairs. Classes used in the specified development samples were excluded; visible-history overlap was permitted. The realized sample contains 4,993 distinct pairs and 3,791 distinct classes. Draws, rather than unique worlds or live model attempts, are the inferential units. Appendix A records the development history, sampling and statistical procedure.

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	Fixed: certificate	Fixed: no certificate	Total
Selected: certificate	282	90	372
Selected: no certificate	1	4,627	4,628
Total	283	4,717	5,000

Table 1: Separate deterministic acquisition study: certificate issuance on 5,000 paired draws, not P/L/U feasibility labels. All 655 issued certificates matched their targets; the 282 jointly certified draws contribute two certificates each. That cell includes 97 draws whose targets were already unique before the observation. The remaining 4,903 draws yield 186 new certificates in the fixed arm and 275 in the selected arm. The net gain is $90 - 1 = 89$ on the primary $N = 5{,}000$. No certificate means unresolved under the corresponding policy, not an impossibility proof for every one-observation method.

All 5,000 pairs completed, with no technical missingness or unknown work counters. The selected policy issued 372 certificates (7.44%), versus 283 (5.66%) for the fixed policy. There were 90 selected-only and one fixed-only certificates: 89 net additional certificates, or **1.78 percentage points**. The pre-specified, at-least-95% confidence interval is **[1.23, 2.29] percentage points**. It excludes zero and supports the narrow positive-difference claim in this frame. It includes the two-percentage-point research-interest threshold, so a gain of at least that size remains unresolved. The planning power calculation is not a guarantee or a post hoc estimate of achieved power. Of the 5,000 draws, 97 already had a unique target before the extra observation and remain in both arms' totals. Among the 4,903 initially ambiguous draws, the observation newly certified 186 targets in the fixed arm and 275 in the selected arm; their net difference is again 89. The primary denominator remains all 5,000 draws.

All 655 issued certificates matched the target audit; any contradiction would have invalidated the certificate result regardless of its interval. Nevertheless, 4,628 selected-policy draws (92.56%) remain uncertified. The one fixed-only certificate also rules out per-record dominance. Query choices differ on 686 draws. Of 458 draws with a pre-observation possible gain, only 90 realize it; the possible-loss draw realizes one loss. These counts separate a favorable candidate branch from the actual response. They are descriptive diagnostics, not further primary tests or calibrated response probabilities.

Equal observation budgets did not imply equal computation. Each arm made 5,000 environment queries and 233,280,000 candidate–history checks. Relative to the fixed arm, the selected policy made 5,283,246 additional query–candidate evaluations and 20,000 selection-branch checks: approximately 59,362 and 225, respectively, per *net* additional certificate. These work categories are reported separately, not added as if they were disjoint CPU instructions. Within-arm reuse is recorded separately and is not counted as additional work. Actor CPU time was 136.38 versus 137.96 seconds; the fixed arm always ran first, so timing is descriptive for this implementation and host. No economic-utility or deployment conclusion is drawn.

This realizes the acquisition benefit bounded by the earlier branch scan. The earlier failed prediction criteria (Appendix A.1) and the separate decision-record crossing results remain unchanged. No general crossing benefit, improved LLM experimentation, universal coverage or cross-contract transfer follows.

Post hoc query comparators on the same inputs. A later reader analysis reconstructed both allowed responses on the original 5,000 draws. Always querying q_1 yields 219 certificates, versus 283 for the pre-specified q_0 and 372 for the selected policy. A policy choosing each query with probability 1/2 independently of the record has an expected $(283 + 219)/2 = 251$ certificates on these inputs. This is an exact expectation, not an observed randomized run. Reconstructed q_0/q_1 certificates have zero target disagreements. These comparisons are retrospective and use no new sample or live observation; they neither replace the primary 89-net gain and its interval nor establish optimality among other selectors. The reader replay computes them from the reconstructed histories, candidate supports and original world tables, separately from historical work counters.

2 Position and scope

Informative query selection is established (Cohn et al., 1994), including identifiability-directed acquisition (Shetty & Braga-Neto, 2026). In active automata learning, Angluin's membership and equivalence-query setting aims to identify a regular language (Angluin, 1987); here the known covering class is a finite transducer class, there is no equivalence oracle, and one of two one-step queries serves a particular future target rather than full system identification. Query by committee uses disagreement to select informative queries (Seung et al., 1992). Our frozen branch-purity selector neither samples a committee nor interprets candidate multiplicities as probabilities; we make no learning-rate comparison to that method. Our contribution is this frozen policy comparison, with realized branches, incorrect-certificate checks and compute accounting. Neither query selection nor unanimity-based inference in a covering finite class is claimed as novel.

The positive average difference is limited to the pre-specified finite frame; new classes, query contracts and live LLM selection remain unassessed. The primary comparison establishes superiority to the frozen fixed q_0 baseline. The q_1 and fair-random expectation above are separate post hoc descriptions, not pre-specified confirmatory tests of a wider optimality claim. The reader replay reconstructs the original sample, histories, query responses, candidate computations and target checks from disclosed synthetic inputs, then recovers the counts and primary interval. It collects no new sample and does not independently attest the historical run. This result and the crossing case study retain separate denominators and claims.

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A Separate finite-class observation-selection study

A.1 Development history and a changed estimand

Earlier prediction criteria remain failed. In one 1,600-draw study, candidate consensus issued 639 point predictions, 189 incorrect (29.58% conditional error). A history filter chosen from those data and tested on a separate 1,600-draw sample issued 436 predictions with 94 errors (21.56%). Both failed the joint target of at most 5% conditional error and at least 25% prediction coverage. The present certificate comparison has a different estimand.

Reanalysis of the second sample with a covering class found 31 initially singleton targets among 1,600 records. A response-branch scan on those records found 134 possible gains and zero possible losses for the selected policy versus the fixed query, but collected no observations. It did not establish realized superiority. The policy was chosen with these results known, then frozen before the new class-frame sample. Development records are excluded from the new denominator.

A.2 Class, target and sampling contract

The candidate class contains deterministic systems with at most three states and binary inputs and outputs, covered by 46,656 labeled three-state tables. Labeled candidates and sampled behavior classes are different units. A certificate requires a nonempty compatible set unanimous about the specified two-output target from the original cutoff. Coverage and implementation correctness are premises, not established for arbitrary worlds by observing no mistakes. Actors receive the legal history, target definition and candidate class, without true system tables or state. The observation tool returns only a permitted intermediate response, never the target continuation.

Write the transition/output table as $(\delta(s,a), \lambda(s,a))$, with $s \in \{0,1,2\}$, $a \in \{0,1\}$ and initial state 0. The table emits $\lambda(s,a)$ on a transition from s and moves to $\delta(s,a)$. The eight observed input–output pairs determine each compatible candidate's cutoff state s_8. Query q_b applies just input b to a fresh copy of that cutoff and returns $\lambda(s_8, b)$, for $b \in \{0,1\}$. The target is

$$T(s_8) = \big(\lambda(\delta(s_8,0),0),\ \lambda(\delta(s_8,1),0)\big).$$

These are the second outputs of $(0,0)$ and $(1,0)$, respectively, starting from separate copies of the original cutoff. The first outputs are not the target; the chosen query does not shift its reference state. The stored target code is $2T_0 + T_1 \in \{0,1,2,3\}$. The fixed arm always chooses q_0; q_0 also wins the selected policy's final tie. Only q_0 and q_1 are legal: a query cannot request the second step or the target pair. Actor-visible definitions and candidate-side computation are disclosed in `artifact/inspection/wire.py` and `compute.py`; they do not supply the true system's table.

Canonicalization gives 11,500 behaviors in the full at-most-three-state class: 4 with one minimal reachable state, 144 with two, and 11,352 with three. The original sampling design restricts worlds to exactly three minimal reachable states, removing the 148 lower-state behaviors. Seven earlier development fixtures represent two distinct eligible three-state behaviors; removing these gives the frozen 11,350-class catalog. The first 1,600-draw development sample uses 1,499 distinct classes, leaving 9,851; the second uses 1,479 additional classes, leaving 8,372. Each has 256 eight-input histories. Thus the apparent 150-class gap is an explicit sampling restriction and prior-fixture exclusion, not a change to the actor's covering hypothesis class. The two classes and all subsequent exclusions are reconstructible by canonical-table identity and index in `artifact/acquisition/replay.py`. The original local structural intent is dated 26 October 2026, before the 27 October main freeze; these local dates and hashes are not independent public time attestations. The fixed sample contains 5,000 draws, uniform with replacement over their 2,143,232 pairs. The implementation uses an operating-system seed, a SHA-256 counter stream and rejection mapping to avoid index-modulo bias. Six upper-residue rejections among 5,006 stream words are sampling implementation events, not outcome exclusions. The inferential model is independent uniform draws from this finite frame, not a claim that cryptographic expansion provides physically independent randomness. All seven duplicate-pair excess draws and 97 initially singleton targets remain in N; 4,903 draws are initially ambiguous. Visible-history overlap is allowed. Class exclusion does not imply absence from catalog construction or all prior research.

A.3 Pre-specified interval, size and decision

Let G and L count selected-only and fixed-only certificates, and $D = G + L$. Write θ for discordance probability and β for the selected-only probability conditional on discordance. The difference in marginal certificate probabilities is

$$\Delta = \theta(2\beta - 1), \qquad \widehat{\Delta} = (G - L)/N.$$

The primary procedure forms central 97.5% Clopper–Pearson intervals for θ from D/N and for β from G/D (full $[0,1]$ when $D = 0$), then takes the minimum and maximum of $\theta(2\beta - 1)$ at their four rectangle corners. Conditional coverage for the second interval, unconditional coverage for the first and the union bound give joint coverage of at least 95% under the paired-draw model. This is a conservative projection, not the same procedure as a marginal Wald interval. The numerical implementation and boundary cases were checked before sampling.

The pre-specified N grid was 1,000 through 6,000 in steps of 1,000. At planning difference 0.02 and discordance 0.15 (selected-only 0.085, fixed-only 0.065), $N = 5{,}000$ was the first to reach the 90% power target:

91.65415%, versus 83.92156% at 4,000. Planning used the same zero-exclusion decision as the primary interval. For positive discordance, it requires the central 97.5% conditional interval for β to exclude one half, rather than substituting a 5% two-sided test. Power depends on the assumed discordance; 0.15 was not a proven upper bound. Two percentage points was a research-interest threshold, not a utility valuation or a claim that the true effect was that large.

One final readout followed all commitments and the target audit. With $G = 90$, $L = 1$, the component intervals are $\theta \in [0.0142214, 0.0229103]$ and $\beta \in [0.9319176, 0.9998618]$. The projected interval is $[0.01228495885, 0.02290400123]$. Its lower endpoint exceeds zero, so the positive-difference claim is supported; its inclusion of 0.02 leaves the interest threshold unresolved. Technical missingness would retain the fixed denominator rather than trigger replacement draws. Any incorrect certificate, identity violation or target leak is a validity failure independent of the confidence interval. There was no outcome-based sample extension or policy revision.

A.4 Execution controls, diagnostics and work

Separate actor processes and state copies implement the two arms. The controls exercise invalid-query rejection, arm-order invariance of normalized scientific outputs, masked-target invariance, rejection of target-continuation access, pre-audit commitments, incorrect-certificate detection and complete work accounting. Five thousand paired predictions were committed before privileged target evaluation. The 10,000 actor processes and keep-awake process exited. Pre-main checks used 77 synthetic extra observations and 81 actor processes; they are excluded from the scientific sample. Same-host checks and clean-copy replay are not independent laboratory validation.

Among initially ambiguous draws, the fixed and selected policies certified 186/4,903 and 275/4,903 targets. Among the 686 draws with different queries, the counts were 31/686 and 120/686. These are descriptive conditional rates. The one fixed-only result means that counting singleton-target branches does not maximize certification for every actual response. Likewise, the 4,627 neither-certified pairs do not prove that every permitted one-observation method would fail.

Measured operation	Fixed	Selected
Environment extra observations	5,000	5,000
Candidate–history checks	233,280,000	233,280,000
Candidate history steps	494,337,048	494,337,048
Query–candidate evaluations	5,283,246	10,566,492
Selection-branch checks	0	20,000
Target–candidate evaluations	5,283,246	5,283,246
Target–candidate steps	21,132,984	21,132,984
Certificate-branch lookups	5,000	5,000
Reused query values within an arm	0	5,283,246
Actor CPU seconds	136.38	137.96
Actor wall seconds	143.91	146.28
Actor-process wall seconds	311.74	313.80

Table 2: Work for the separate 5,000-draw acquisition study. Counters are operation-specific and may overlap; reused values are not additional evaluations. No cross-arm or cross-record caching is used. Timing does not identify an order-independent speed difference.

The total main-run elapsed time was 640.28 seconds. In addition to 10,000 extra observations, environment accounting records 40,000 history-generation steps, 5,000 resets, 10,000 clones, zero history-replay steps and 20,000 target-audit steps. The audit compares 655 issued certificates to their targets. These operation-specific counters are not an additive cost score. Timing is descriptive because the fixed arm always ran first. Model/provider requests were zero; research-assistant use was unmeasured.

A.5 Raw-input reader reconstruction

The bundle includes the original catalogs, exclusions, seed, rejection receipts, frozen plan, commitments, events, histories and arm records, plus collector/environment source as inspection-only text. Plain JSON columns and string references retain every original field and reproduce all 5,000 original record hashes. Retained process IDs and timings establish byte identity only.

Run `python3.12 -I -B -S artifact/acquisition/replay.py` from the bundle root. It reconstructs the universe, ordered catalogs, exclusions and full first development manifest; verifies the original hashes and 5,006 sampling receipts; regenerates histories, both policies' query responses, candidate computations and certificates; and fixes predictions before recomputing targets. Every scientific field and work counter is compared before recomputing the primary interval. The archived commitment-before-truth event order is also checked.

This standard-library replay collects no new sample, calls no model, and executes neither participant programs nor the historical isolation collector. Independent historical attestation remains outside its scope.